

Lab program-35:

35. Consider a file system that brings all the file pointers together into an index block. The *i*th entry in the index block points to the *i*th block of the file. Design a C program to simulate the file allocation strategy.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_BLOCKS 100

struct Block {
    int blockNumber;
    char data[256];
};

int main(void) {
    FILE *file;
    struct Block block;
    int blockNumber;
    long indexBlock[MAX_BLOCKS];

    for (int i = 0; i < MAX_BLOCKS; i++)
        indexBlock[i] = -1;

    file = fopen("indexed_file.dat", "wb");
    if (file == NULL) {
        perror("Error opening the file");
        return 1;
    }

    printf("Enter blocks (Enter '0' as block number to exit):\n");
    while (1) {
        printf("Block Number: ");
        scanf("%d", &block.blockNumber);
        if (block.blockNumber == 0)
            break;
        if (block.blockNumber < 1 || block.blockNumber >= MAX_BLOCKS) {
            printf("Invalid block number.\n");
            continue;
        }

        printf("Data: ");
        scanf("%255[^\n]", block.data);
        indexBlock[block.blockNumber] = ftell(file);
```

```

        indexBlock[block.blockNumber] = fgetc(file);
        fwrite(&block, sizeof(block), 1, file);
    }
    fclose(file);

    file = fopen("indexed_file.dat", "rb");
    if (file == NULL) {
        perror("Error opening the file");
        return 1;
    }

    while (1) {
        printf("Enter the block number to read (0 to exit): ");
        scanf("%d", &blockNumber);
        if (blockNumber == 0)
            break;
        if (blockNumber < 1 || blockNumber >= MAX_BLOCKS || indexBlock[blockNumber] == -1) {
            printf("Block not found.\n");
            continue;
        }

        fseek(file, indexBlock[blockNumber], SEEK_SET);
        if (fread(&block, sizeof(block), 1, file) == 1) {
            printf("Block Number: %d\n", block.blockNumber);
            printf("Data: %s\n", block.data);
        }
    }

    fclose(file);
    return 0;
}

```

OUTPUT:

Enter blocks (Enter '0' as block number to exit):

Block Number: 39

Data: JSAWNTH

Block Number: 43

Data: SAI

Block Number: 12

Data: FRIEND

Block Number: 0

Enter the block number to read (0 to exit): 12

Block Number: 12

Data: FRIEND

Enter the block number to read (0 to exit): 0

Process exited after 34.15 seconds with return value 0
Press any key to continue . . . |